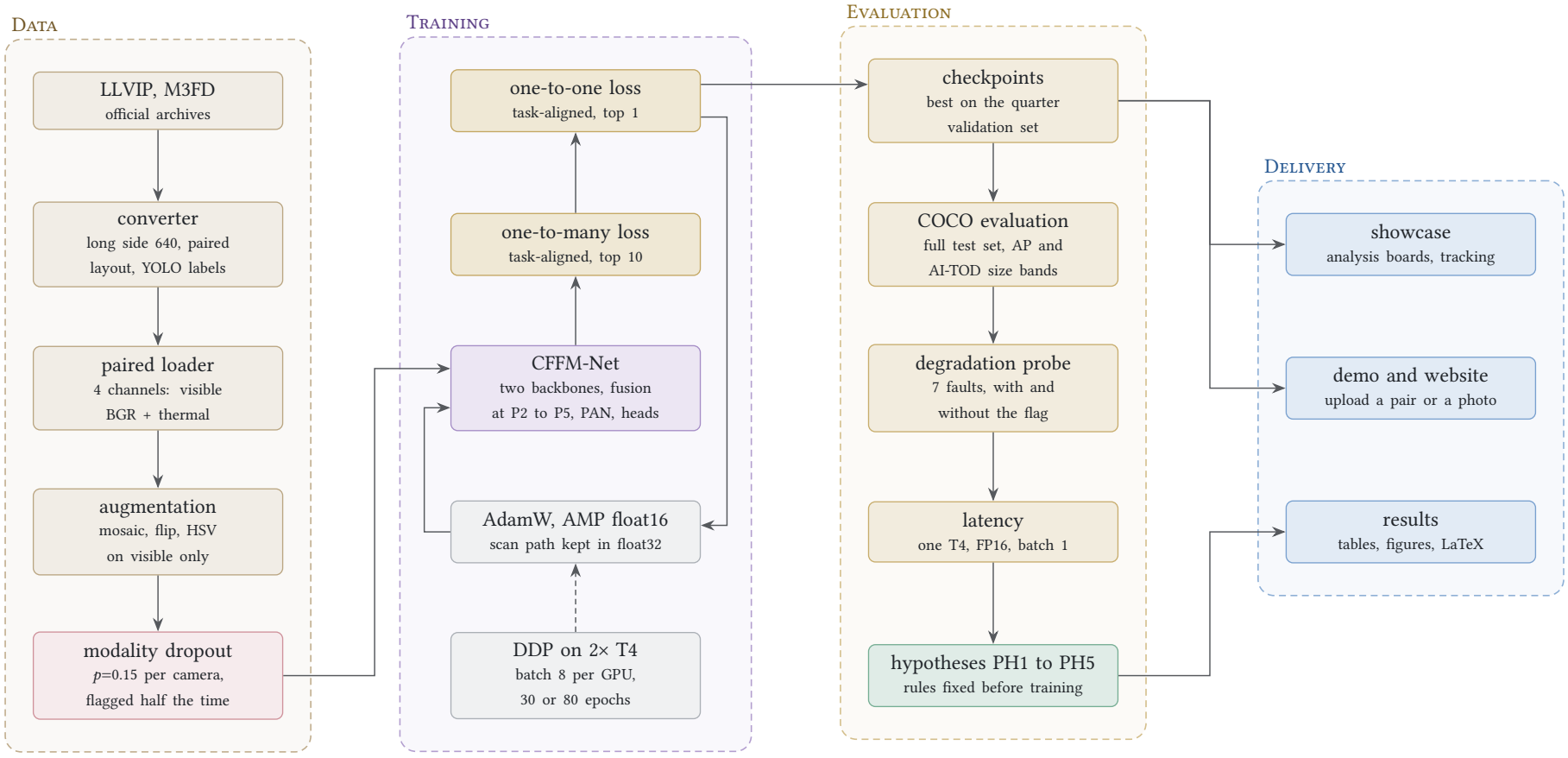
Data
LLVIP, M3FD
official archives
converter
long side 640, paired layout, YOLO labels
paired loader
4 channels: visible BGR + thermal
augmentation
mosaic, flip, HSV on visible only
modality dropout
p=0.15 per camera, flagged half the time
Training
one-to-one loss
task-aligned, top 1
one-to-many loss
task-aligned, top 10
CFFM-Net
two backbones, fusion at P2 to P5, PAN, heads
AdamW, AMP float16
scan path kept in float32
DDP on 2× T4
batch 8 per GPU, 30 or 80 epochs
Evaluation
checkpoints
best on the quarter validation set
COCO evaluation
full test set, AP and AI-TOD size bands
degradation probe
7 faults, with and without the flag
latency
one T4, FP16, batch 1
hypotheses PH1 to PH5
rules fixed before training
Delivery
showcase
analysis boards, tracking
demo and website
upload a pair or a photo
results
tables, figures, LaTeX